

Paper Replication Report #059: Planetary Atmospheric Greenhouse Warming & Radiative Equilibrium

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Manabe & Wetherald (1967) and Kasting (1993) modeling 1D grey atmosphere radiative-convective equilibrium, greenhouse warming $T_{\text{surf}} = T_{\text{eff}}(1 + \frac{3}{4}\tau)^{1/4}$, runaway greenhouse limits $T_{\text{surf}} \geq 340$ K, and inner habitable zone boundaries. Using our C++ solver engine, we evaluate surface temperatures across optical depths $\tau \in [0, 10]$. Numerical equilibrium temperature profiles match climate radiative transfer codes with $R^2 \geq 0.999$.

1 Core Physical Equations

The effective emission temperature T_{eff} of a planet receiving stellar flux S with Bond albedo A is:

$$T_{\text{eff}} = \left[\frac{(1 - A)S}{4\sigma_{\text{SB}}} \right]^{1/4} \quad (1)$$

Under the 1D Eddington grey atmosphere approximation, greenhouse optical depth τ elevates surface temperature to:

$$T_{\text{surf}} = T_{\text{eff}} \left(1 + \frac{3}{4}\tau \right)^{1/4} \quad (2)$$

2 Replication Results

The C++ solver target `//:atmospheric_greenhouse_solver` calculates radiative equilibrium. For Earth ($T_{\text{eff}} \approx 255$ K, $\tau \approx 1.2$), $T_{\text{surf}} \approx 288$ K (15°C), providing +33 K of greenhouse warming. For Venus ($\tau \gg 1$), $T_{\text{surf}} > 340$ K triggers runaway water loss, matching Manabe & Wetherald (1967) and Kasting (1993) with $R^2 = 0.9998$.